Explanation sheet for the dataset: Participant survey data from the citizen science project FLOW, 2021

For details on research questions, study design and methods, see the following publication:
von Gönner, J., Masson, T., Köhler, S., Fritsche, I., Bonn, A. (in press): Citizen science promotes knowledge, skills and collective action to monitor and protect freshwater streams. *People and Nature.*

Overview of variables

No: Number of respondent
ID: Anonymous respondent identification code to match individual responses in pretest, posttest and follow-up.
group: Respondents affiliation to one of the three groups: Treatment (citizen scientists), Control1 (waiting control group) and Control2 (panel control group).
complete: Number of surveys completed by each respondent (2 = pretest and posttest; 3 = pretest, posttest and follow-up)
age: Age of respondents in years
gender: Respondents' gender (male or female)

attitude.1: Mean score of respondents' self-reported attitudes towards stream protection at pretest (for scale items see questionnaire below, no. 7).
attitude.2: Mean score of respondents' self-reported attitudes towards stream protection at posttest (see below, question no. 7).
attitude.3: Mean score of respondents' self-reported attitudes towards stream protection at follow-up test (see below, question no. 7).

awareness.1: Mean score of respondents' self-reported awareness of stream stressors towards at pretest (see below, question no. 9).
awareness.2: Mean score of respondents' self-reported awareness of stream stressors towards at posttest (see below, question no. 9).
awareness.3: Mean score of respondents' self-reported awareness of stream stressors towards at follow-up test (see below, question no. 9).

behavioral_control.1: Mean score of respondents' self-reported behavioural control regarding stream protection at pretest (see below, question no. 6).
behavioral_control.2: Mean score of respondents' self-reported behavioural control regarding stream protection at posttest (see below, question no. 6).
behavioral_control.3: Mean score of respondents' self-reported behavioural control regarding stream protection at follow-up test (see below, question no. 6).

collective_efficacy.1: Mean score of respondents' self-reported collective efficacy for stream protection at pretest (see below, question no. 8).
collective_efficacy.2: Mean score of respondents' self-reported collective efficacy for stream protection at posttest (see below, question no. 8).
collective_efficacy.3: Mean score of respondents' self-reported collective efficacy for stream protection at follow-up test (see below, question no. 8).

group_id.1: Mean score of respondents' self-reported group identification at pretest (see below, question no. 10).
group_id.2: Mean score of respondents' self-reported group identification at posttest (see below, question no. 10).
group_id.3: Mean score of respondents' self-reported group identification at follow-up test (see below, question no. 10).

Overview of variables (continued)

group_norms.1: Mean score of respondents' perceived social norms at pretest (see below, question no. 10).
group_norms.2: Mean score of respondents' perceived social norms at posttest (see below, question no. 10).

intrinsic_motivation: Mean score of respondents' self-reported intrinsic motivation to participate in citizen science stream monitoring (see below, no. 24).

knowledge_score.1: Mean score of respondents' measured stream-related knowledge at pretest (see below, question no. 17).
knowledge_score.2: Mean score of respondents' measured stream-related knowledge at posttest (see below, question no. 17).
knowledge_score.3: Mean score of respondents' measured stream-related knowledge at follow-up test (see below, question no. 17).

nature_rel.1: Mean score of respondents' self-reported nature relatedness at pretest (see below, question no. 12).
nature_rel.2: Mean score of respondents' self-reported nature relatedness at posttest (see below, question no. 12).
nature_rel.3: Mean score of respondents' self-reported nature relatedness at follow-up test (see below, question no. 12).

interest.1: Mean score of respondents' self-reported interest in biology and in streams at pretest (see below, question no. 14).
interest.2: Mean score of respondents' self-reported interest in biology and in streams at posttest (see below, question no. 14).

past_behavior1.1: Mean score of respondents' self-reported personal behaviour to protect streams at pretest (see below, question no. 4).
past_behavior1.2: Mean score of respondents' self-reported personal behaviour to protect streams at posttest (see below, question no. 4).
past_behavior1.3: Mean score of respondents' self-reported personal behaviour to protect streams at follow-up test (see below, question no. 4).

past_behavior2.1: Mean score of respondents' self-reported collective behaviour to protect streams at pretest (see below, question no. 4).
past_behavior2.2: Mean score of respondents' self-reported personal behaviour to protect streams at posttest (see below, question no. 4).
past_behavior2.3: Mean score of respondents' self-reported personal behaviour to protect streams at follow-up test (see below, question no. 4).

plan_behavior1.1: Mean score of respondents' self-reported personal action intentions to protect streams at pretest (see below, question no. 5).
plan_behavior1.2: Mean score of respondents' self-reported personal action intentions to protect streams at posttest (see below, question no. 5).
plan_behavior1.3: Mean score of respondents' self-reported personal action intentions to protect streams at follow-up test (see below, question no. 5).

plan_behavior2.1: Mean score of respondents' self-reported collective action intentions to protect streams at pretest (see below, question no. 5).
plan_behavior2.2: Mean score of respondents' self-reported collective action intentions to protect streams at posttest (see below, question no. 5).
plan_behavior2.3: Mean score of respondents' self-reported collective action intentions to protect streams at follow-up test (see below, question no. 5).

Overview of variables (continued)

personal_efficacy.1: Mean score of respondents' self-reported personal efficacy to protect streams at pretest (see below, question no. 8).
personal_efficacy.2: Mean score of respondents' self-reported personal efficacy to protect streams at posttest (see below, question no. 8).
personal_efficacy.3: Mean score of respondents' self-reported personal efficacy to protect streams at follow-up test (see below, question no. 8).

personal_norms.1: Mean score of respondents' self-reported personal norms regarding stream protection at pretest (see below, question no. 8).
personal_norms.2: Mean score of respondents' self-reported personal norms regarding stream protection at posttest (see below, question no. 8).
personal_norms.3: Mean score of respondents' self-reported personal norms regarding stream protection at follow-up test (see below, question no. 8).

skills.1: Mean score of respondents' self-reported skills for stream monitoring at pretest (see below, question no. 16).
skills.2: Mean score of respondents' self-reported skills for stream monitoring at pretest (see below, question no. 16).
skills.3: Mean score of respondents' self-reported skills for stream monitoring at pretest (see below, question no. 16).

Explanation sheet for the dataset: Participant survey data from the citizen science project FLOW, 2021

Overview of survey questions and items. The survey questions were translated from the original language German to English for this overview. Column four provides information on measurement times for each question. Questions marked with (*) were only included in the treatment group surveys. Greyed out questions are not included in the analysis and datasheet for the current study.

No.	Topic/ Construct	Question and items	Measurement time
1	**Visiting rivers & streams**	**How often have you spent time by a river or stream in the last 3 weeks?** ***Please choose one answer.*** □ not at all □ once or twice □ about once a week □ several times a week □ (almost) every day	Pretest, Posttest
2	**Cultural ecosystem services**	**Why do you spend time by rivers or streams?** ***You can choose up to five answers.*** □ to go for a walk □ to do sports □ to go on a boat trip □ to go fishing □ to go swimming □ to spend time in nature □ to watch animals □ to find peace and quiet □ to find shade and cooling in summer □ to meet other people □ to get from place A to place B □ to record species □ to carry out maintenance and protection measures (e.g. collecting rubbish) □ to learn something new □ other (please specify) □ I don't usually spend time near streams or rivers	Pretest, Posttest
3	**Ideas to protect streams**	**Imagine that you are asked to help protect a stream in your area.** ***Please select one answer.*** □ I have ideas about what I could do to help protect the stream. □ I don't know what I could contribute. *If you have ticked answer 1, please let us know how you could help to protect the stream. Please enter your answer in bullet points. [Open text field]*	Pretest, Posttest
4	**Personal and collective behaviour** (items 1-4: personal behaviour, items 5-8: collective behaviour)	***Please indicate how often you have done the following things in the last 3 weeks.*** **[Follow-up test:** ***Please indicate how often you have done the following things in the last 6 weeks.*****]** (1)=not at all, (2)=rarely, (3)=sometimes, (4)=often, (5)=very often □ respect rules of water and river protection areas (e.g. not leaving litter behind, not entering closed areas) □ buy organic food □ use insecticides or herbicides □ use biodegradable detergents for cleaning and washing □ learn about water and stream protection by attending events or lectures □ talk to family or friends about water and stream protection (in person or via social media) □ participate in group-based events to protect rivers and streams (beyond the scope of the CS project FLOW, e.g. collecting rubbish or planting trees along river banks) □ contact local politicians or environmental agencies with own ideas for stream protection	Pretest, Posttest, Follow-up

Explanation sheet for the dataset: Participant survey data from the citizen science project FLOW, 2021

Overview of survey questions and items (continued)

No.	Topic/ Construct	Question and items	Measurement time
5	**Personal and collective action intentions** (items 1-4: personal behaviour, items 5-8: collective behaviour)	**Do you plan to do anything to protect rivers and streams in the next 3 weeks? [Follow-up test: Do you plan to do anything to protect rivers and streams in the next 6 weeks?]** ***Please tick one answer for each item.*** (1)=does not apply, (2)=rather does not apply, (3)=applies partly, (4)=rather applies, (5)=applies completely □ respect rules of water and river protection areas (e.g. not leaving litter behind, not entering closed areas) □ buy organic food □ use insecticides or herbicides □ use biodegradable detergents for cleaning and washing □ learn about water and stream protection by attending events or lectures □ talk to family or friends about water and stream protection (in person or via social media) □ participate in group-based events to protect rivers and streams (beyond the scope of the CS project FLOW, e.g. collecting rubbish or planting trees on river banks) □ contact local politicians or environmental agencies with own ideas for stream protection	Pretest, Posttest, Follow-up
6	**Perceived behavioural control**	***Please select one answer to rate the two following statements:*** □ How easy is it for you personally to do concrete things in your everyday life to protect rivers and streams (e.g. not leaving litter behind, buying organic food or using biodegradable detergents)? □ How easy is it for you to participate in group-based events for stream protection with people from your community (e.g. collecting rubbish or planting trees along river banks)? (1)=very difficult, (2)=rather difficult, (3)=undecided, (4)=rather easy, (5)=very easy	Pretest, Posttest, Follow-up
7	**Attitudes towards stream protection measures**	**The following measures all contribute to water and stream protection in different ways. How useful do you personally think it is to implement these measures to protect rivers and streams?** ***Please tick one answer for each item.*** (1)=not useful at all, (2)=rather not useful, (3)=undecided, (4)=rather useful, (5)=very useful □ buying organic food □ not using insecticides or weed killers □ using biodegradable cleaning and washing detergents □ taking part in group-based events to protect streams (e.g. collecting rubbish or planting trees on river banks) □ participate in the monitoring of aquatic species □ contact local environmental agencies with own ideas for stream protection	Pretest, Posttest, Follow-up

Explanation sheet for the dataset: Participant survey data from the citizen science project FLOW, 2021

Overview of survey questions and items (continued)

No.	Topic/ Construct	Question and items	Measurement time
8	**Personal norms, personal and collective efficacy**	**To what extent do you agree with the following statements on water and stream protection?** ***Please select one answer for each item.*** *Note: The term "ecological status" refers to the water quality of a stream and the living conditions it offers for plants and animals.* (1)=strongly disagree, (2)=rather disagree, (3)=partly agree, 4=rather agree, (5)=strongly agree □ I believe that together with other people in my community, I can improve the ecological status of rivers and streams. □ Because of my values, I want to do something to protect rivers and streams. □ I believe that my personal actions can help protect rivers and streams. □ I feel responsible to take action to protect rivers and streams. □ It is important to me to do something about the pollution of rivers and streams. □ I would feel guilty if I didn't do something to protect rivers and streams.	Pretest, Posttest, Follow-up
9	**Awareness of stream stressors**	**How threatening do you think the following things are for river and stream ecosystems?** ***Please tick one answer for each item.*** (1)=not threatening at all, (2)=rather not threatening, (3)=partly threatening, (4)=rather threatening, (5)=very threatening □ Agricultural use of floodplains (direct environment of rivers and streams) □ People walking along river banks □ Stabilization of river banks with concrete or stones □ Pesticide inputs into rivers or streams from nearby fields and gardens □ Organic farming with at least 5 meters distance from the river bank □ Straightening of the watercourse	Pretest, Posttest, Follow-up
10	**Group identification, perceived social norms** (in citizen science project group)	**To what extent do you agree with the following statements?** ***Please select one answer for each item.*** (1)=strongly disagree, (2)=rather disagree, (3)=partly agree, 4=rather agree, (5)= strongly agree, (-1)=I don't know. □ I feel very connected to the members of my FLOW project group. □ Most of the members of my FLOW project group want to engage in stream protection. □ Most of the members of my FLOW project group are active in stream protection beyond the immediate scope of the citizen science project.	*Question only for citizen science group at pretest, posttest (and at follow-up for first item on group identification)
11	**Simple activities in nature** (see Richardson et al. 2020)	***Please indicate how often you have done the following things in the last 3 weeks.*** **[Follow-up test:** ***Please indicate how often you have done the following things in the last 6 weeks.*****]** (1)=not at all, (2)=rarely, (3)=sometimes, (4)=often, (5)=very often □ spend time relaxing in a beautiful place in nature □ take photos or draw landscapes, plants or animals □ closely observe wild animals such as birds, insects or fish □ consciously listen to nature sounds (e.g. bird song, rushing water) □ closely observe the ecological status of streams as you pass them	Pretest, Posttest, Follow-up

Explanation sheet for the dataset: Participant survey data from the citizen science project FLOW, 2021

Overview of survey questions and items (continued)

No.	Topic/ Construct	Question and items	Measurement time
12	**Nature relatedness** (NR-6 scale, Nisbet & Zelenski 2013)	**To what extent do you agree with the following statements?** ***Please select one answer for each item.*** (1)=strongly disagree, (2)=rather disagree, (3)=partly agree, 4=rather agree, (5)= strongly agree □ My ideal vacation spot would be a remote, wilderness area. □ I always think about how my actions affect the environment. □ My connection to nature and the environment is a part of my spirituality. □ I take notice of wildlife wherever I am. □ My relationship to nature is an important part of who I am. □ I feel very connected to all living things and the earth.	Pretest, Posttest, Follow-up
13	**Understanding of science, Trust in science**	***Please select one answer:*** **When I hear the term "scientific study," I have...** □ a clear idea of what it means. □ a general idea of what it means. □ little knowledge of what it means. ***Please select one answer:*** **How much do you trust in science?** (1)=don't trust at all, (2)=rather don't trust, (3)=trust partly, (4)=rather trust, (5)=trust completely, (-1)=I don't know.	Pretest, Posttest (+ Follow-up for question 1 on understanding of science)
14	**Interest in biology and in streams**	***Please indicate to which extent you are interested in the following things:*** (1)=not at all, (2)=a little, (3)=partially, (4)=a lot, (5)=very strongly □ Biology □ Ecological research □ Rivers and stream □ Freshwater protection □ Environmental policy	Pretest, Posttest
15	**Prior experience**	***Please indicate how often you have done the following things before participating in the citizen science project FLOW:*** (1)=not yet at all, (2)=once, (3)=several times (2-3x), (4)=often (4-10x), (5)= very often (more than 10x) □ collect data on the occurrence of plant or animal species □ assess the ecological status of a river or stream	* Question only for citizen science group at pretest
16	**Self-reported skills**	***A) Please rate your skills in the following disciplines:*** (1)=none at all, (2)=very little, (3)=moderate, (4)=good, (5)= very good □ Ecology in general □ Stream ecology ***B) Please rate the following statements:*** **I currently have the abilities to…** 1)=strongly disagree, (2)=rather disagree, (3)=partly agree, 4=rather agree, (5)= strongly agree □ understand an ecological field protocol □ collect standardized data on the ecological status of streams □ assess the ecological status of streams based on my own citizen science data	Pretest, Posttest, Follow-up

Explanation sheet for the dataset: Participant survey data from the citizen science project FLOW, 2021

Overview of survey questions and items (continued)

No.	Topic/ Construct	Question and items	Measurement time
16	**Self-reported skills**	***C)*** **How would you rate your capability to identify macroinvertebrates (assuming that you have an illustrated identification booklet and stereo microscope with 20-fold magnification to help you)?** ***Please choose one answer.*** □ 1- Very low: I can distinguish insects from worm-like animals, mussels and snails. However, I do not yet know the different orders of macroinvertebrates. □ 2- Low: I can recognize some common macroinvertebrate orders (e.g. caddisfly larvae, mayfly larvae, dragonfly larvae). □ 3- Solid: I can recognize most native macroinvertebrate orders and correctly identify some common macroinvertebrate taxa to the family level (e.g. gammarids, isopods). □ 4 - Good: I can distinguish the different macroinvertebrate orders without problems and I'm able to identify several taxa to the family level. I immediately recognize common taxa such as *Gammarus*, *Ephemera*, *Asellus*, *Erpobdella*. □ 5- Very good: I can correctly identify many taxa down to the family or even genus level. I immediately recognize characteristic taxa such as *Baetis*, *Dugesia*, *Calopteryx*, *Heptageniidae*, *Limnephilidae*.	Pretest, Posttest, Follow-up
17	**Objective knowledge**	Multiple choice question 1: **What are the three main criteria of ecological river and stream assessment according to the European Water Framework Directive?** ***Up to three answers are possible.*** □ Hydromorphology □ Algae community □ Proportion of agricultural land use in catchment area □ Physico-chemical water quality □ Hydrology □ Stabilization of river banks □ Straightening of river banks □ Riverbed morphology □ Biological communities □ Fish fauna □ Flow diversity □ Pollutant concentration □ I don't know Multiple choice question 2: **What are bioindicators?** ***You can select up to two answers.*** □ common species that disappear when the ecosystem is affected by a lack of oxygen □ species whose presence indicates certain environmental factors □ important for ecology because they can be used to accurately measure environmental factors □ sensitive species that require high oxygen saturation and high solar radiation for survival □ common species that are found in almost all habitats □ species that are particularly sensitive to changes in their habitat □ I don't know Open question 3: **What proportion of streams and rivers in Germany are in good ecological condition?** ***Please enter a percentage. If you don't know, please estimate _____*** ****Note:*** *To calculate a knowledge score for each respondent, we summed the results for the three questions. For multiple choice questions 1 and 2, we counted 1 point for each correct answer and 0 points for each incorrect answer. For question 3, respondents could earn a maximum of 2 points. This resulted in a total range of 0 to 7, which we recoded to 1 to 8 for adjustment to the other scales.*	Pretest, Posttest, Follow-up

Explanation sheet for the dataset: Participant survey data from the citizen science project FLOW, 2021

Overview of survey questions and items (continued)

No.	Topic/ Construct	Question and items	Measurement time
18	**Age**	**How old are you?** ***Please enter your age in years*** _____	Pretest
19	**Gender**	**What is your gender?** ***Please choose one answer.*** □ female □ male □ diverse	Pretest
20	**Group affiliation**	**Do you belong to any of the following organizations?** ***Please choose one answer.*** □ BUND (Friends of the Earth Germany) □ NABU (Nature And Biodiversity Conservation Union Germany) □ Angling club □ High school □ University or research institute □ Other organization (please specify) □ I don't belong to any of these organizations.	Pretest
21	**Education**	**What is your highest level of formal education?** ***Please select one answer.*** □ Secondary school diploma □ High school diploma/ A-level □ Vocational training □ University degree (Bachelor) □ University degree (Master) □ Doctorate degree □ Other (please specify)	Pretest
22	**Work area**	**Do you work in any of the following areas?** ***Please select one answer.*** □ Science □ Environmental protection □ Media / communication □ Education □ Government, policy making □ Law and administration □ Agriculture, forestry or horticulture □ No, I am a high school or university student or trainee. □ No, I work in another area	Pretest
23	**Voluntary engagement**	**Are you engaged in any volunteer work in your free time?** ***Please select one answer.*** □ yes □ no □ in planning **If you ticked 'yes' or 'in planning', in what area are you involved or planning to get involved?** ***Multiple answers are possible.*** □ Extracurricular educational work with youth or adults □ Professional advocacy outside the workplace □ Leisure and social activities □ Health care □ Justice and crime problems □ Church and religion □ Culture and music □ Politics □ School and kindergarten □ Social work □ Sports and exercise □ Environment, nature conservation or animal welfare □ Rescue service or volunteer fire department □ Other area (please specify)	Pretest
24	**Motivation to participate** (adapted from Phillips, Porticella & Bonney 2017: Motivation to Participate in Citizen Science Scale)	**Why did you decide to participate in the citizen science project FLOW?** ***Please select one answer for each item.*** (1)=strongly disagree, (2)=rather disagree, (3)=partially agree, (4)=rather agree, (5)=strongly agree □ because I enjoy outdoor activities □ because my engagement is appreciated by others □ because I think the approach of citizen science is a good idea □ because the citizen science project makes me feel part of an active community □ because I want to support environmental research □ because other people would be disappointed if I did not participate □ because I want to learn how research works □ because I would feel guilty if I didn't get involved in protecting the environment □ because the project topic is very important to me	Posttest *Items on intrinsic motivation that were analysed for this study are marked in black, items on extrinsic motivation are marked in grey.*

Explanation sheet for the dataset: Participant survey data from the citizen science project FLOW, 2021

Overview of survey questions and items (continued)

No.	Topic/ Construct	Question and items	Measurement time
25	**Fieldwork team**	**Which team did you mainly work in during the stream monitoring event?** ***Please select one answer.*** □ Hydromorphology team □ Water chemistry team □ Macrozoobenthos team	Posttest
26	**Stream status**	**In what ecological status was the stream you examined on the day of your citizen science monitoring event?** ***Please select one answer.*** □ very good (near-natural condition, large variety of macroinvertebrates) □ good (near-natural condition, human-made changes to stream morphology visible in places, many different macroinvertebrates present) □ moderate (human-made degradation of stream hydromorphology clearly visible in several places, diversity of macroinvertebrates limited) □ poor (rather unnatural condition, clear degradation of the ecosystem due to construction measures or agricultural activities visible in numerous places, only few macroinvertebrate species present) □ bad (poor condition, heavily degraded by construction measures or agricultural activities, very few macroinvertebrate species present)	Posttest
27	**Usefulness of CS results**	***Please rate the following statements.*** **In my opinion, the results of the citizen science stream assessment are...** (1)=strongly disagree, (2)=rather disagree, (3)=partly agree, 4=rather agree, (5)= strongly agree □ well comprehensible and understandable data □ data that should be critically examined □ useful data that can be compared to results of other FLOW monitoring events □ scientifically usable data □ a good basis for implementing stream protection or restoration measures	Posttest
28	**General feedback on CS project FLOW**	**Do you have feedback for us regarding the FLOW citizen science project communication, training or monitoring event?** ***Please enter your feedback in bullet points. [open text field]***	Posttest